Kempe-Based 5-Colour-Theorem Colouring Program:

March 7, 2026

A program that 5-colors any **simple, planar uncoloured graph**.

Original Kempe proof (without counterexample):

Since every planar graph has a node with degree < 6 (5-degenerate), removing a node with degree < 6 forms an induced planar subgraph also possessing a node with degree < 6. So, using Kempe's method for colouring nodes with degree < 6, we can colour all nodes recursively in the reverse order of deletion until the entire graph is coloured.

Proof by Induction:

P(n): every simple, planar graph with n vertices is 4-colorable.

Base case:

P (1) is true since a single node can be coloured with 1 colour, 1 < 4.

Inductive step:

Inductive hypothesis- Assume P(k) is true for some k.

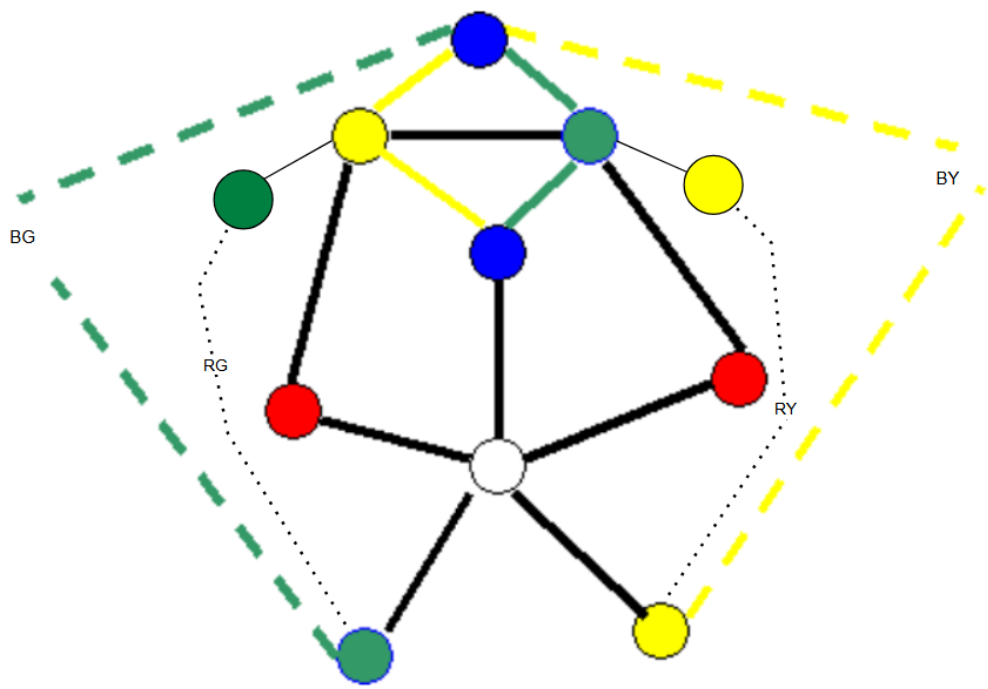
Then add an uncoloured node to the graph to check P(k+1). If the added node has degree < 6, apply Kempe's method and 4 colour the graph.

If the added node has a degree >= 6, remove a coloured node with <6 neighbours, by 5-degeneracy this node must exists, and by the inductive hypothesis P(k), the induced subgraph is 4 colourable. Then add the deleted node back in (as an uncoloured node) and apply Kempe's method to 4-colour the graph.

So, P(k+1) is true.

By mathematical induction, Kempe's method with fact of 5-degeneracy of planar graphs proves the four-colour theorem.

Our Counterexample: Kempe's method for the 4-colour theorem fails at the following counterexample:



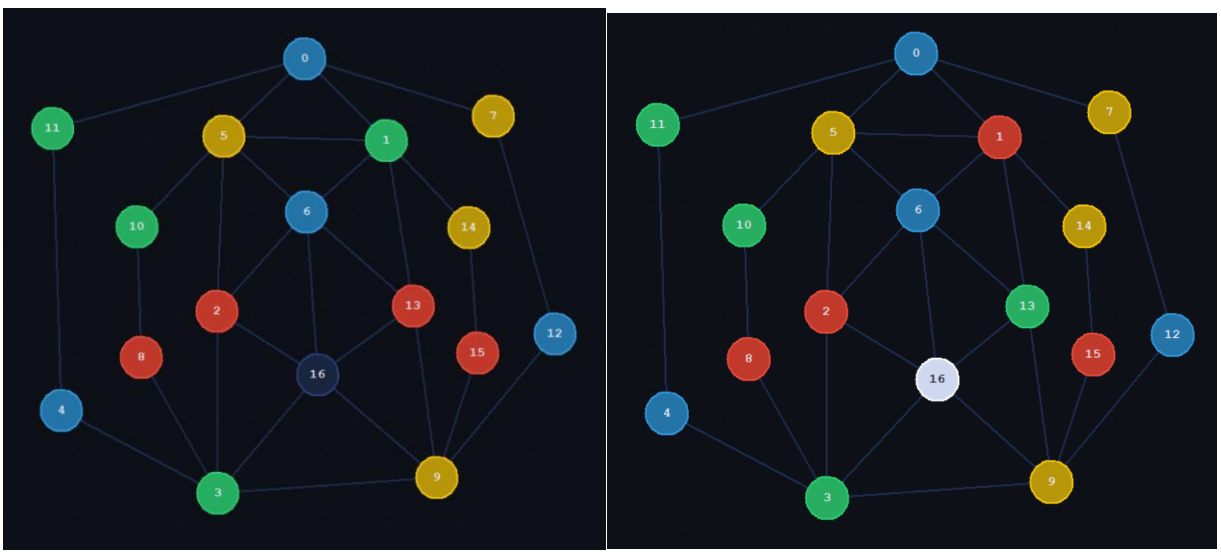
5 Colouring Program -O(n^2):

In our program, we colour the uncoloured node WHITE for the counterexample to Kempe's method, applying a five-colour theorem.

The **input** to the program should be a <u>simple, uncoloured, planar</u> graph adjacency matrix and the number of nodes (precompile). The program will work if a K5 minor exists due to 5CT implementation but will fail for a K6 minor. The **output** is the colouring of the graph in nodal order.

Test Cases:

<u>Applying the code to our counterexample produces the following:</u>

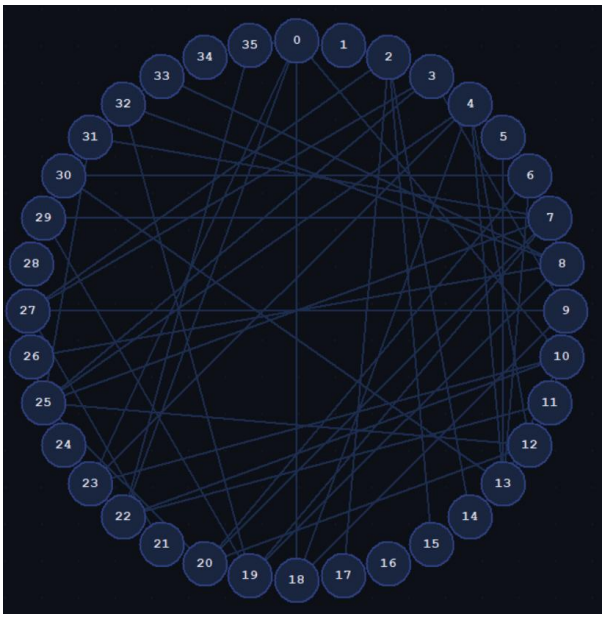


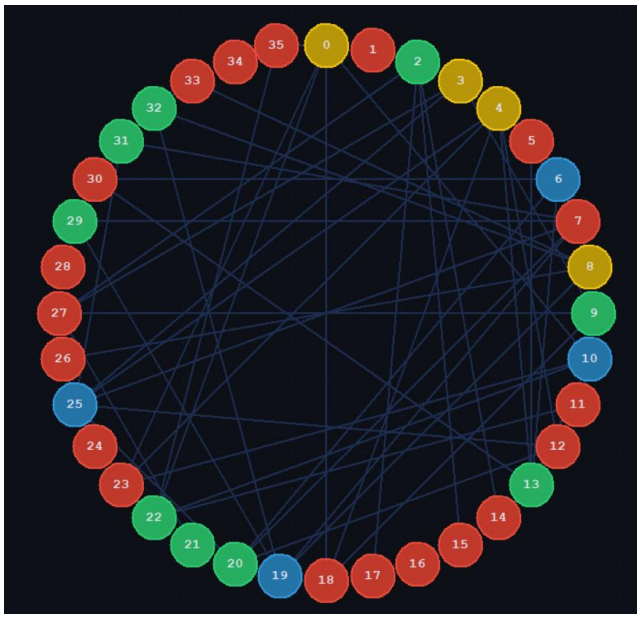
Note: Applying the program to an already coloured graph, such as our counterexample, requires a specific set up in the code to show.

<u>K5:</u> the program colours the graph with 5 colours: R, G, B, Y, W.

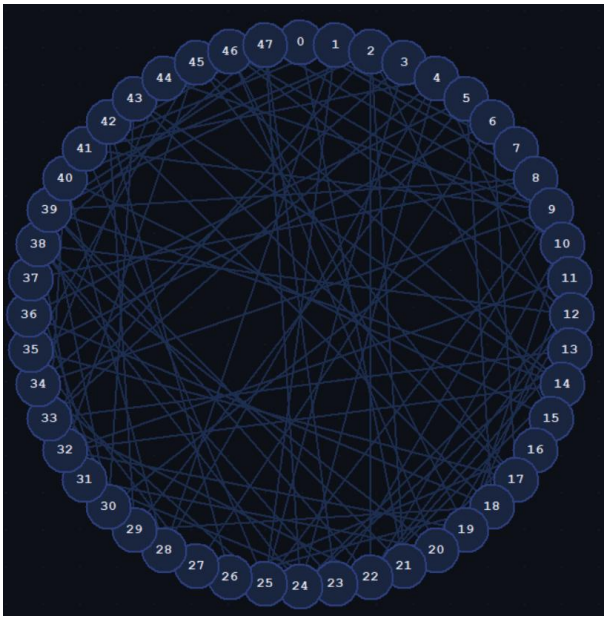
<u>Various countries with complex provinces, states, governorates, oblasts were used. Areas were in alphabetical order; hence they can be adjacent.</u>

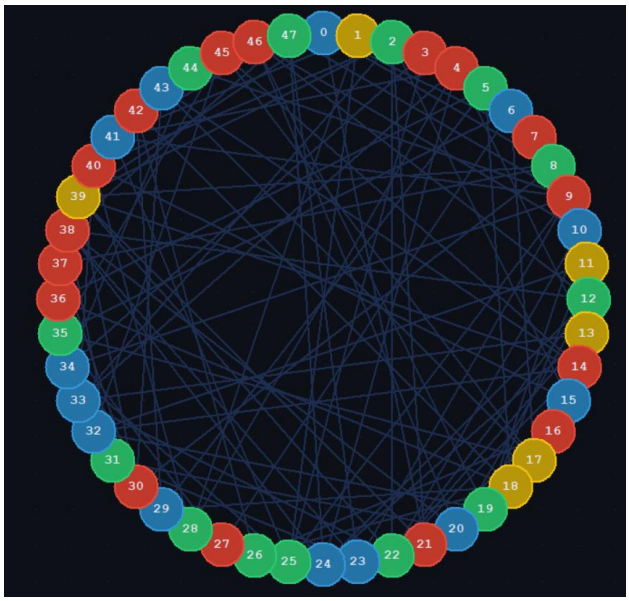
India (unconnected graph):



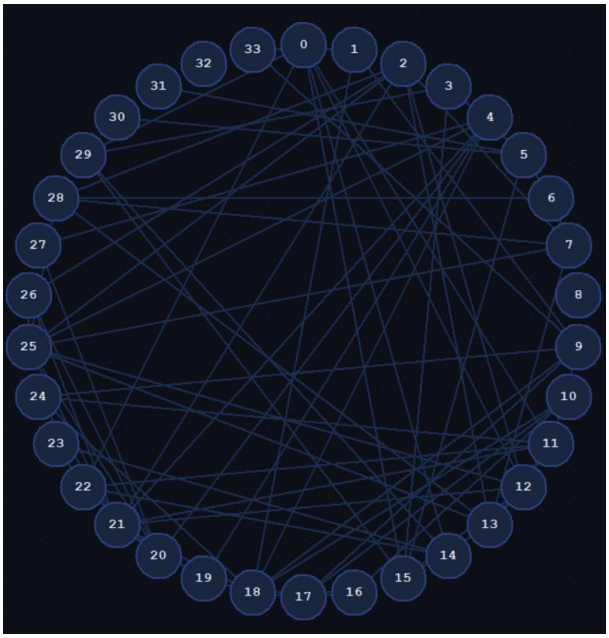


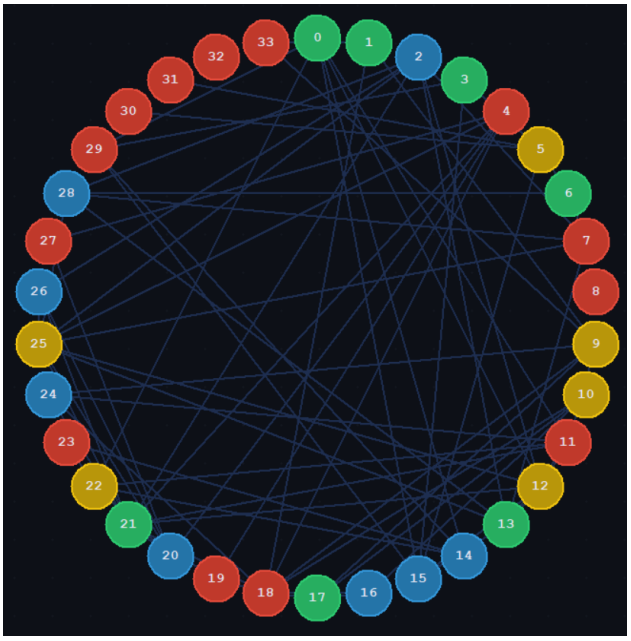
USA (connected graph, omitted alaska hawaii):



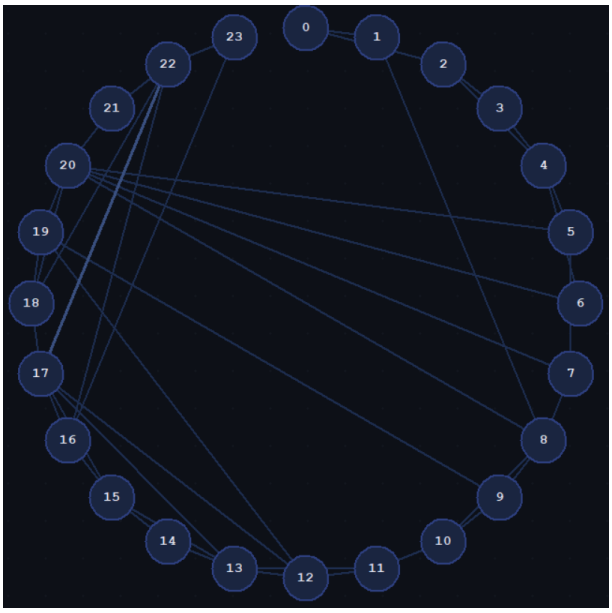


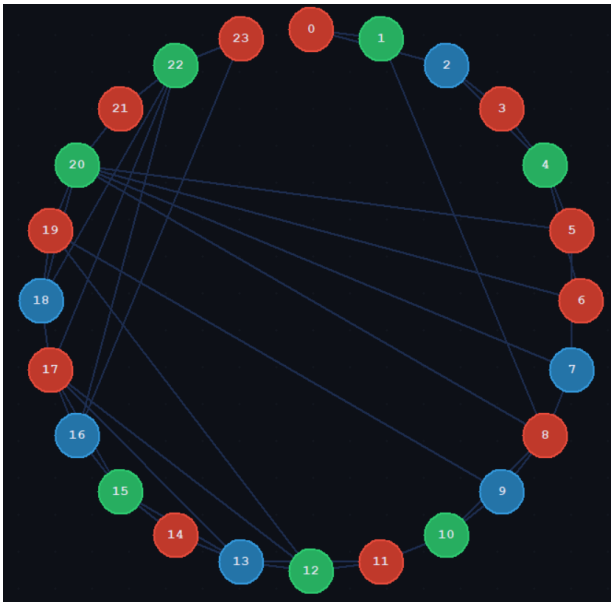
China:



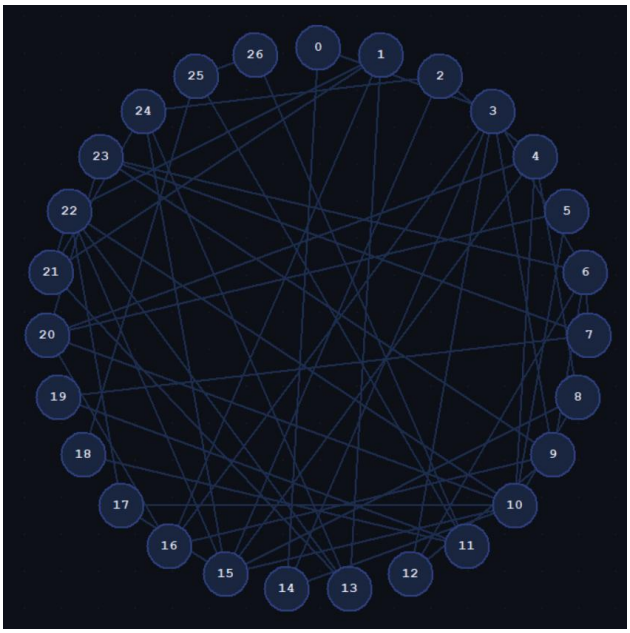


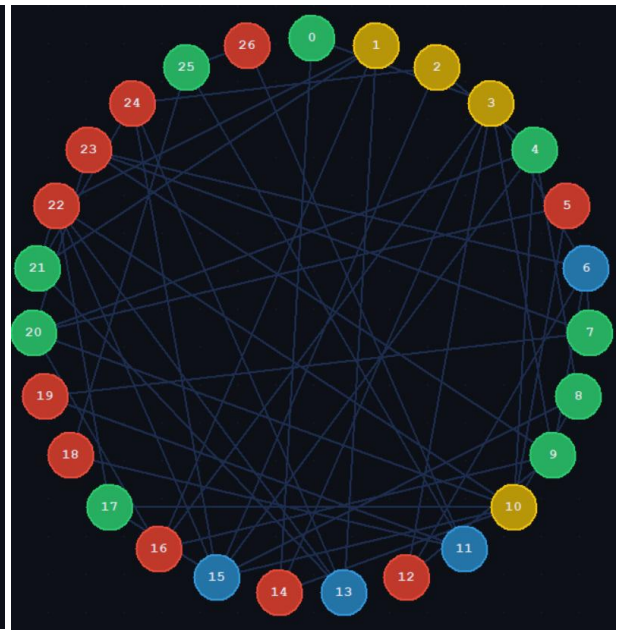
Ukraine:





Egypt:





Next: Kempe 2.0

Can try test case of Kn,n,n,n,n

For map of USA- Use Graphviewer:

USA graph IMPORT:

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